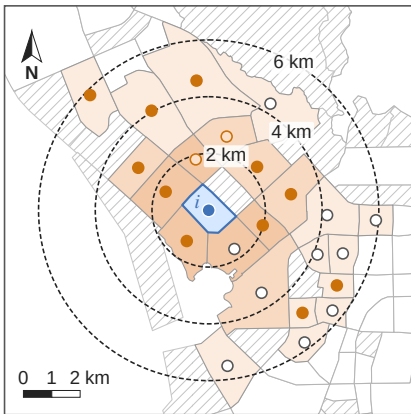


Distance-ring spillover exposure



$$S_{i,t}^{(k)} = \text{mean}\{\Delta\ell_{j,t} : j \in \mathcal{R}_k(i)\}, \quad \mathcal{R}_k(i) = \{j \neq i : r_{k-1} \leq d_{ij} < r_k\}, \quad d_{ij}: \text{centroid distance}$$

$$\eta_{i,t}^{\text{spill}} = \sum_k \delta_k S_{i,t}^{(k)}; \quad \text{an empty ring gives } S_{i,t}^{(k)} = 0$$